BEING INTEGRAL IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A morphism is integral if and only if it is affine and universally closed. Morphisms, Lemma morphisms-lemma-integral-universally-closed, invoked at proof lines 6310-6312. Splits integrality into two properties.
- Universal closedness and affineness are fpqc local on the base. Same chapter, Lemmas lemma-descending-property-universally-closed and lemma-descending-property-affine, invoked at proof lines 6313-6315. Gives both ingredients, hence the result.

Source reference. Original lemma 02L5: sources/stacks-descent/pinned-original-source.tex, source lines 6143–6143. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KS: sources/stacks-descent/pinned-original-source.tex, source lines 5732–5732. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WM: sources/stacks-chow/tags.txt, source lines 2487–2487. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is integral" is fpqc local on the base.*

Proof. An integral morphism is the same thing as an affine, universally closed morphism. See Morphisms, Lemma 01WM. Hence the lemma follows on combining Lemmas 02KS and 02L5. □

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